

Urban Power Grid Resilience Assessment Method Considering Urban Lifeline Function

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Abstract—The occurrence probability of extreme disasters such as earthquakes is very low, but when they do occur, they pose significant risks to the stable and reliable operation of power systems. During the Wenchuan Earthquake in China on May 12, 2008, with a magnitude of 8.0, the power grid experienced a load loss exceeding 4 million kW, leaving at least 2.46 million users without electricity, resulting in enormous economic losses and severe social impacts. As a crucial energy provider in urban systems, the power grid plays a key supporting role in the operation of urban lifeline systems. Therefore, resilience assessment of urban power grids should not be limited to power restoration but should comprehensively consider the recovery of functional demands across urban lifeline systems. This study establishes a coupled system of water, electricity, and gas networks as urban lifelines and proposes urban power grid resilience assessment metrics based on the functional demands of urban lifeline systems. These metrics take into account the needs of different types of users for water, electricity, and gas resources under conditions where main grid power supply is constrained by earthquake disasters, focusing on voltage levels ranging from 10 kV to 220 kV.

Keywords—power grid resilience; urban lifeline; earthquake disaster; urban power grid

I. INTRODUCTION

The importance of electricity in the national economy is becoming increasingly prominent, playing an irreplaceable role in people's lives and social development. Therefore, higher requirements are imposed on the operation of power systems: the power system must exhibit high reliability in the face of known threats with traditionally high confidence levels, while also demonstrating sufficient resilience when dealing with sudden high-impact low-probability (HILP) events[1]. Since the

concept of resilience was first introduced by Holling in 1973, it has been widely applied across various disciplines, including engineering and public administration[2]. In electrical engineering, the definition of power grid resilience varies among organizations and researchers. However, its core lies in the ability of the grid to withstand and recover from major disturbance events, including the capability of power systems to predict, absorb, resist, respond to, and quickly recover from interruptions caused by HILP events [3].

As a typical HILP event, earthquakes have posed significant risks to the stable and reliable operation of power systems in recent years. During the Wenchuan Earthquake in China on May 12, 2008, with a magnitude of 8.0, at least 1,700 10 kV lines, 106 35 kV lines, 118 110 kV lines, and 46 220 kV lines were out of service. The total load loss exceeded 4 million kW, and at least 2.46 million users experienced power outages. The direct economic loss surpassed 10.6 billion RMB, causing enormous economic damage and severe social impacts [4]. Urban systems, as an integrated whole, rely on the coordination of various lifeline systems, such as power grids, water supply networks, and gas supply networks. While these systems perform their respective functions, they are also coupled. The power grid, as the energy provider within urban systems, plays a key supporting role in the operation of urban lifeline systems. Any grid failure directly affects the operational status of other systems[5]. In recent years, resilience assessment of power systems under seismic disasters has become a research hotspot. For instance, reference[6] quantitatively analyzed the power restoration process for hospitals under seismic disasters, considering the influence of different lifeline systems on functional recovery. Reference[7] constructed resilience assessment metrics for island integrated energy systems from the perspectives of

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robustness, rapidity, and redundancy, fully evaluating the power supply capability of island systems under natural disasters such as earthquakes. Reference [8] proposed a three-stage robust optimization model to minimize load curtailment caused by natural disasters. Reference [9] leveraged vehicle-mounted mobile energy storage systems and electric vehicles as grid support resources to enhance the resilience of distribution systems during disasters.

However, the resilience assessment of power grids under disasters, considering the functional demands of various systems, requires further research. In Chapter 2 of this paper, seismic attenuation models are used to obtain peak ground acceleration (PGA) parameters for each unit. The PGA parameters are then employed to derive component failure probability models under different seismic intensities, generating failure scenarios. Chapter 3 proposes urban power grid resilience recovery metrics based on the functional demands of urban lifeline systems to evaluate grid resilience under seismic disasters. Chapter 4 establishes models for water network operations coupled with energy systems, gas network operations coupled with energy systems, and main distribution network operations. Chapter 5 provides example scenarios and case studies. Chapter 6 analyzes the case study results and presents conclusions, verifying the reliability of the proposed metrics.

II. SEISMIC DISASTER MODEL AND COMPONENT FAILURE PROBABILITY MODEL

A. Seismic Disaster Model

Currently, peak ground acceleration (PGA) is primarily used as the basis for seismic resistance in engineering design. Using a magnitude of 6.5 as the threshold, the seismic magnitude is divided into segments to construct the attenuation patterns of ground motion parameters for different magnitude ranges:

$$M \leq 6.5$$

$$\lg(Y_{PGA}) = 0.561 + 0.746M - 1.925 \cdot \lg(L + 0.956 \cdot \exp(0.462 \cdot M)) \quad (1)$$

$$M \geq 6.5$$

$$\lg(Y_{PGA}) = 2.501 + 0.448M - 1.925 \cdot \lg(L + 0.956 \cdot \exp(0.462 \cdot M)) \quad (2)$$

Where, M represents the earthquake magnitude, L represents the epicentral distance, Y_{PGA} represents the peak ground acceleration of ground motion.

B. Power Grid Line Failure Probability Model

In the face of earthquakes of varying magnitudes, damage can be categorized into: minor damage, moderate damage, severe damage, and complete failure. Vulnerability models are commonly used to describe the relationship between the severity of damage and the failure probability of power grid lines under different earthquake magnitudes. These models are typically derived through empirical analysis of historical data or theoretical studies. At present, the vulnerability curves of components are widely described using the cumulative log-normal distribution function:

$$P_j^k(Y_{PGA}) = \int_0^{Y_{PGA}} \frac{1}{\sqrt{2\pi}\xi_j^k x} \exp\left[-\frac{(\ln x - \ln \varepsilon_j^k)^2}{2(\xi_j^k)^2}\right] dx \quad (3)$$

Where, $P_j^k(Y_{PGA})$ represents the probability of component k being in the j -th damage state under a peak ground acceleration (PGA) of Y_{PGA} ; $j = 1 \sim 4$ denotes the four damage states: minor damage, moderate damage, severe damage, and complete failure; ε_j^k and ξ_j^k represent the median and standard deviation of the vulnerability curve for the four damage states, respectively.

This study only considers the damage to the urban lifeline system's functionality caused by power grid disruptions under seismic disasters. Therefore, it is assumed that the gas network and water network can maintain basic operations.

III. URBAN POWER GRID RESILIENCE RECOVERY METRICS BASED ON URBAN LIFELINE SYSTEM FUNCTIONAL DEMANDS

The objective of this study is to minimize the functional loss of the entire urban lifeline system under seismic disaster scenarios by considering the resource demands of various users for electricity, water, and gas.

$$F = \min \sum_{t=1}^T \sum_{i=1}^{\Omega} \omega_i \left(\alpha_i \cdot \frac{P_{i,t}^{De} - P_{i,t}^{Re}}{P_{i,t}^{De}} + \beta_i \cdot \frac{W_{i,t}^{De} - W_{i,t}^{Re}}{W_{i,t}^{De}} + \gamma_i \cdot \frac{G_{i,t}^{De} - G_{i,t}^{Re}}{G_{i,t}^{De}} \right) \quad (4)$$

Where, ω_i represents the importance weight of user i ; $\alpha_i, \beta_i, \gamma_i$ denote the demand coefficients for electricity, water, and gas resources in the energy structure of user i respectively, for the same user i , $\alpha_i, \beta_i, \gamma_i$ vary based on their functionality. The more reliant a user is on a particular resource, the larger the corresponding coefficient value; $P_{i,t}^{Re}, W_{i,t}^{Re}, G_{i,t}^{Re}$ represent the restoration amounts of electricity, water, and gas resources for user i during time period t under disaster scenarios; $P_{i,t}^{De}, W_{i,t}^{De}, G_{i,t}^{De}$ denote the demand amounts of electricity, water, and gas resources for user i during time period t under normal conditions.

IV. OPERATIONAL AND ENERGY COUPLING MODEL OF URBAN LIFELINE SYSTEMS

The functional relationship between urban lifeline systems and the energy supplied by the power grid is not a simple linear relationship. Moreover, the internal energy demands of various urban lifeline systems are not merely additive. Therefore, it is necessary to consider the differences in resource demand among different types of users and construct an energy coupling model for urban lifeline systems.

The power flow model of the power system can be referenced from [10], the water supply network flow model from [11], and the gas network flow model from [12]. This study focuses on modeling the operational and energy coupling of water supply networks and gas networks.

A. Water Supply Network Energy Coupling System Model

In the water network, pump units serve as electricity-consuming components, forming the coupling link between the power grid and the water network. Therefore, it is necessary to construct an energy coupling model based on the operational characteristics of the pumps.

1) Pump unit operational state model

$$\begin{cases} \Delta h_{jk}^t = -(\alpha W_{jk}^t + \beta) & u_{pump}^t = 1 \\ W_{jk}^t = 0 & u_{pump}^t = 0 \end{cases} \quad (5)$$

$$\forall j \rightarrow k \in \Omega_m^{\text{pipe}}, m \in \Omega_{\text{pump}}$$

Where, u_{pump}^t is a binary variable representing the on/off state of the pump; it takes the value 1 when the pump is running and 0 otherwise; α, β are characteristic parameters of the pump; Ω_m^{pipe} represents the set of pipelines influenced by the operational state of pump m ; Ω_{pump} denotes the set of all pumps.

2) Energy coupling model of pump units

$$P_{m,pump}^t = \frac{\rho g}{\eta_m^{\text{pump}}} (\alpha W_{jk}^t + \beta (W_{jk}^t)^2) \quad (6)$$

$$\forall j \rightarrow k \in \Omega_m^{\text{pipe}}, m \in \Omega_{\text{pump}}$$

Where, $P_{m,pump}^t$ is the electricity consumption of pump m at time t ; ρ is the density of water; g represent the gravitational acceleration; η_m^{pump} is the power factor of pump m ; W_{jk}^t is the flow rate in the pipeline where pump m is located; $P_{m,pump}$ is the power upper limit of pump unit m .

B. Energy Coupling System Model of Gas Supply Network

In the gas supply network, the energy coupling units include compressors and gas turbines. Compressors consume electrical energy, while gas turbines, on the other hand, consume natural gas resources while supplying electrical energy to the power grid. Together, these two units form a bidirectional flow of electricity and gas resources.

1) Compressor operational state model

$$\begin{cases} p_j^t \leq p_k^t \leq \varphi p_j^t & u_{\text{comp}}^t = 1 \\ G_{jk}^t = 0 & u_{\text{comp}}^t = 0 \end{cases} \quad (7)$$

$$\forall j \rightarrow k \in \Omega_n^{\text{pipe}}, n \in \Omega_{\text{comp}}, \forall t \in \Omega_T$$

Where, u_{comp}^t represents a binary variable representing the on/off state of the compressor; it takes the value 1 when the compressor is operating and 0 otherwise; φ is the compression ratio of the compressor; Ω_n^{pipe} represents the set of pipelines influenced by the operational state of compressor n ; Ω_{comp} is the set of all compressors.

2) Energy coupling model of compressors and gas turbines

$$P_{n,comp}^t = \tau_n^{\text{comp}} \cdot G_{jk}^t \quad (8)$$

$$\forall j \rightarrow k \in \Omega_n^{\text{pipe}}, n \in \Omega_{\text{comp}}, \forall t \in \Omega_T$$

$$P_{l,GT}^t = \eta_l^{\text{GT}} \cdot GT_l^t \quad \forall l \in \Omega_{\text{GT}}, \forall t \in \Omega_T \quad (9)$$

Where, P_n^{comp} represent the power consumed by compressor n at time t ; τ represent the conversion coefficient between gas flow rate and power consumption; $P_{l,GT}^t$ represent the electricity generated by gas turbine l at time t ; η_l^{GT} represent the conversion coefficient between gas consumption and electricity generation.

V. CASE STUDY

This study employs the RTS-24 model, the IEEE 33-bus model, the 20-node gas supply network model, and the 15-node water supply network model to simulate the coupling relationships within the urban lifeline system. In this case study, there are a total of 3 pump units, 2 compressor units, 2 gas turbine units, and 2 distributed wind power units. The IEEE 33-bus model includes 5 available tie lines, which are normally open under standard conditions, ensuring the distribution network remains radial, without considering looped network scenarios.

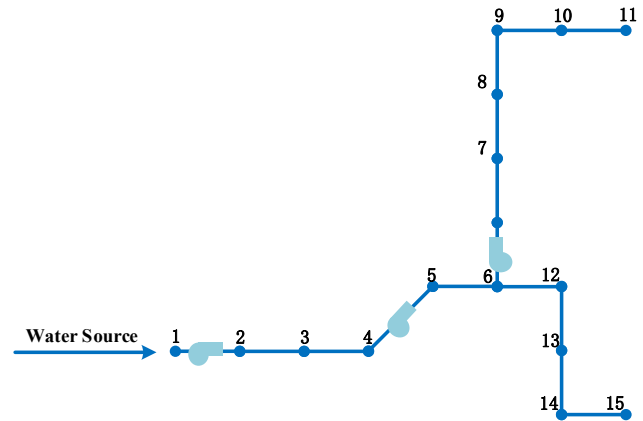


Fig. 1. 15-Node Water Supply Network Model.

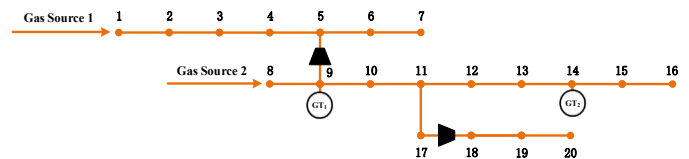


Fig. 2. 20-Node Gas Supply Network Model.

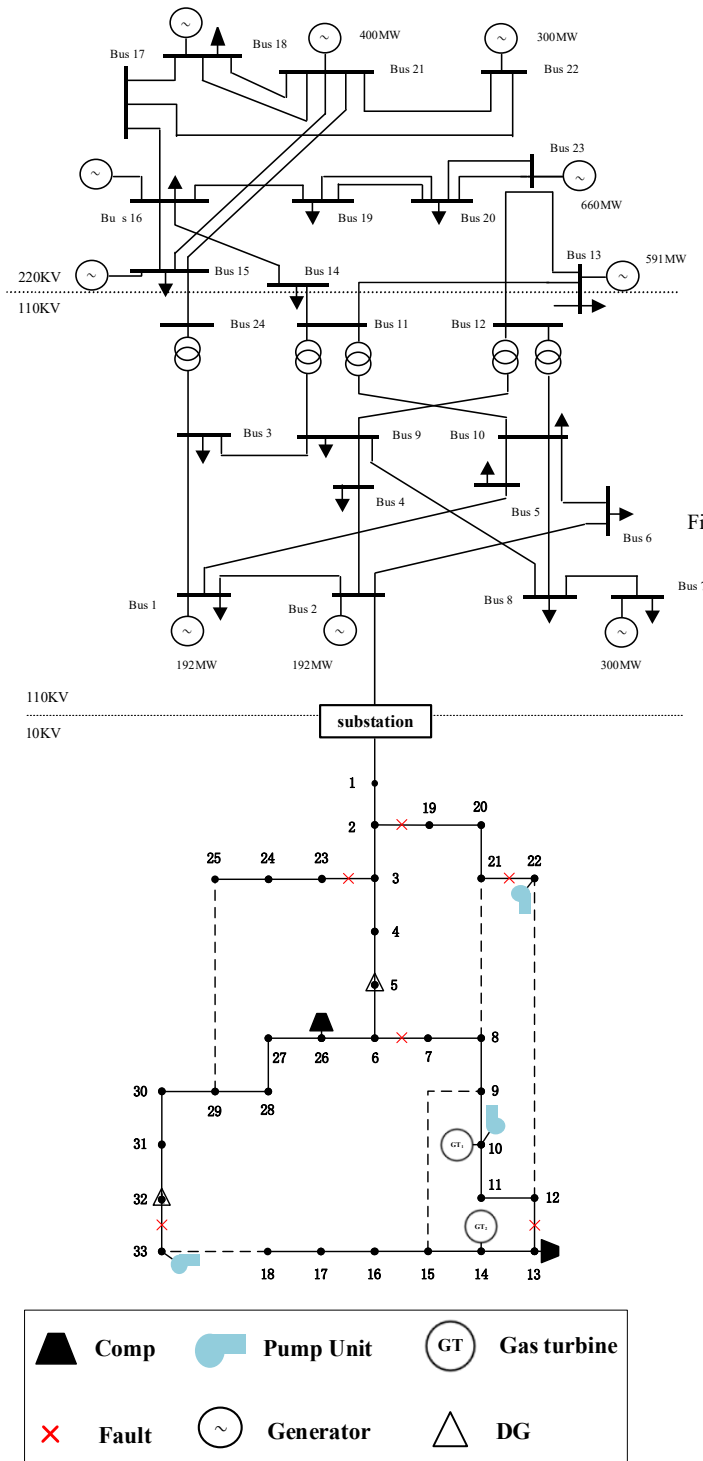


Fig. 3. Example Diagram of Collaborative Operation Between Main and Distribution Networks.

Figures 1 to 3 respectively illustrate the topological structures of the water network, gas network and power grid in the case study. The validity of the proposed metrics is verified by setting two objectives: minimizing the functional loss of the entire urban lifeline system under seismic scenarios and minimizing the power grid's load shedding.

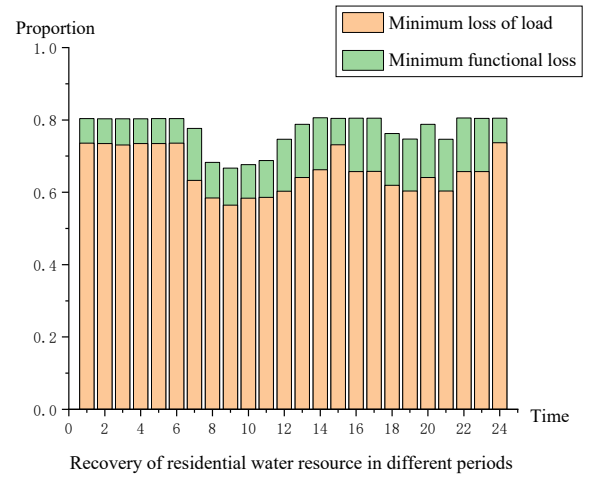


Fig. 4. 24-Hour Comparison of Water Resource Restoration for Critical Users.

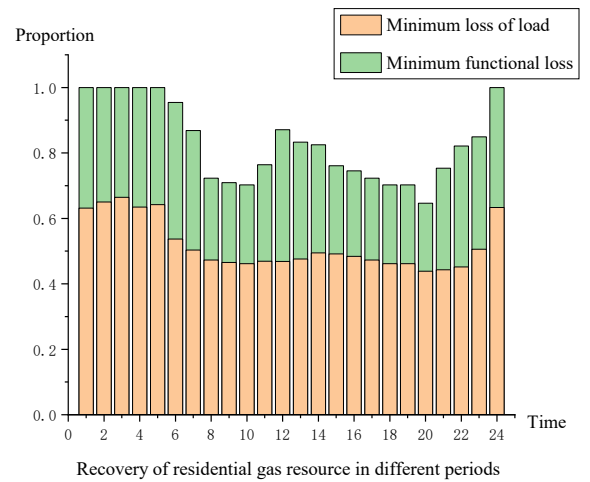


Fig. 5. 24-Hour Comparison of Gas Resource Restoration for Critical Users.

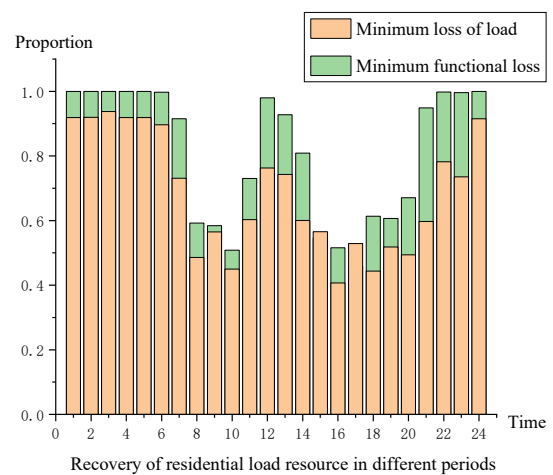


Fig. 6. 24-Hour Comparison of Load Restoration for Critical Users.

Figures 4 to 6 illustrate the 24-hour restoration progress of water, gas, and electricity resources for critical users. It can be observed that, under the objective of minimizing functional loss, power resources are better allocated. When the main grid supply is constrained, water, gas, and electricity resources achieve a higher overall level of restoration. This improvement is particularly significant in the gas supply network with bidirectional energy flow, which not only meets the demand for power restoration but also flexibly allocates gas resources for power generation, thereby ensuring a higher degree of natural gas availability for residential use.

VI. CONCLUSION

The case study results demonstrate that when power resources are constrained, and the power grid experiences line faults, the evaluation method based on the resilience recovery metrics of urban power grids, considering the functional demands of urban lifeline systems, can more effectively guide the allocation of power resources. Both power load restoration and the recovery of water and gas resources show overall improvement. Furthermore, this approach, based on the functional demands of urban lifeline systems, facilitates more effective coordination among various energy networks, such as the interaction between electrical and gas resources. This evaluation method uses power load as a pivot point to balance the demand relationships among the three networks, better meeting the functional demands of various components within the city.

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